

```
import pandas as pd
import re
```

```
#loading a transaction dataset
transaction_data = pd.read_excel("QVI_transaction_data.xlsx")
transaction_data
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PR
0	43390	1	1000	1	5	Natural Chip Compny SeaSalt175g CCs Nacho Cheese 175g	
1	43599	1	1307	348	66	Smiths Crinkle Cut Chips Chicken 170g	
2	43605	1	1343	383	61	Smiths Chip Thinly S/Cream&Onion 175g	
3	43329	2	2373	974	69	Kettle Tortilla ChpsHny&Jlpno Chili 150g	
4	43330	2	2426	1038	108	...	
...	...	...	...	...	...	Kettle Sweet Chilli And Sour Cream 175g	
264831	43533	272	272319	270088	89	Tostitos Splash Of Lime 175g	
264832	43325	272	272358	270154	74	Doritos Mexicana 170g	
264833	43410	272	272379	270187	51	Doritos Corn Chip Mexican Jalapeno 150g	
264834	43461	272	272379	270188	42	Tostitos Splash Of Lime 175g	
264835	43365	272	272380	270189	74		

264836 rows × 8 columns

```
#performing EDA on transaction dataset
transaction_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264836 entries, 0 to 264835
Data columns (total 8 columns):
```

```
# Dtype----- 264836 non-null int64
--- 264836 non-null int64-----
0 DATE_CARD_NBR 264836 non-null int64
1 STORE_NBR 264836 non-null int64 264836 non-null
2 int64 264836 non-null object
3 TXN_ID 264836 non-null int64
4 PROD_NBR 264836 non-null float64
5 PROD_NAME
6 PROD_QTY
7 TOT_SALES
```

dtypes: float64(1), int64(6), object(1)
memory usage: 16.2+ MB

```
transaction_data.describe()
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR
count	43464.036260	135.08011	1.355495e+05	1.351583e+05	56.583157
mean	105.389282	76.78418	8.057998e+04	7.813303e+04	32.826638
std	43282.000000	1.00000	1.000000e+03	1.000000e+00	1.000000
min	43373.000000	70.00000	7.002100e+04	6.760150e+04	28.000000
25%	43464.000000	130.00000	1.303575e+05	1.351375e+05	56.000000
50%	43555.000000	203.00000	2.030942e+05	2.027012e+05	85.000000
75%	43646.000000	272.00000	2.373711e+06	2.415841e+06	114.000000
max					

```
transaction_data.isnull().sum()
```

	0
DATE	0
STORE_NBR	0
LYLTY_CARD_NBR	0
TXN_ID	0
PROD_NBR	0
PROD_NAME	0
PROD_QTY	0
TOT_SALES	

dtype: int64

```
transaction_data.count()
```

	264836 ⁰
DATE	264836
STORE_NBR	264836
LYLTY_CARD_NBR	264836
TXN_ID	264836
PROD_NBR	264836
PROD_NAME	264836
PROD_QTY	264836
TOT_SALES	

dtype: int64

```
#Loading purchase behaviour dataset
purchase_behaviour_data = pd.read_csv("QVI_purchase_behaviour.csv")
purchase_behaviour_data
```

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER
	1000	YOUNG SINGLES/COUPLES	Premium
0	1002	YOUNG SINGLES/COUPLES	Mainstream
1	1003	YOUNG FAMILIES	Budget
2	1004	OLDER SINGLES/COUPLES	Mainstream
3	1005	MIDAGE SINGLES/COUPLES	Mainstream
4		...	...
...	...	MIDAGE SINGLES/COUPLES	Mainstream
72632	2370651	YOUNG FAMILIES	Mainstream
72633	2370701	YOUNG FAMILIES	Premium
72634	2370751	OLDER FAMILIES	Budget
72635	2370961	YOUNG SINGLES/COUPLES	Mainstream
72636	2373711		

72637 rows × 3 columns

Next steps:

Generate code with purchase_behaviour_data

New interactive sheet

```
#performing EDA on purchase behaviour dataset
purchase_behaviour_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 72637 entries, 0 to 72636
Data columns (total 3 columns):
#   Column                Non-Null Count  Dtype
---  -
0   LYLTY_CARD_NBR         72637 non-null  int64
1   LIFESTAGE              72637 non-null  object
2   PREMIUM_CUSTOMER      72637 non-null  object
dtypes: int64(1), object(2)
memory usage: 1.7+ MB
```

```
purchase_behaviour_data.describe()
```

	LYLTY_CARD_NBR
count	1.361859e+05
mean	8.989293e+04
std	1.000000e+03
min	6.620200e+04
25%	1.340400e+05
50%	2.033750e+05
75%	2.373711e+06
max	

```
purchase_behaviour_data.count()
```

	72637
LYLTY_CARD_NBR	72637
LIFESTAGE	72637
PREMIUM_CUSTOMER	

dtype: int64

```
transaction_data.shape
```

```
(264836, 8)
```

```
purchase_behaviour_data.shape
```

```
(72637, 3)
```

```
transaction_data.columns
```

```
Index(['DATE', 'STORE_NBR', 'LYLTY_CARD_NBR', 'TXN_ID', 'PROD_NBR',
      'PROD_NAME', 'PROD_QTY', 'TOT_SALES'],
      dtype='object')
```

```
purchase_behaviour_data.columns
```

```
Index(['LYLTY_CARD_NBR', 'LIFESTAGE', 'PREMIUM_CUSTOMER'], dtype='object')
```

```
transaction_data.duplicated().sum()
#one rows is identited as duplicated
```

```
np.int64(1)
```

```
#verifying duplicate values
duplicate_rows = transaction_data[transaction_data.duplicated(keep=False)]
duplicate_rows
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_Q
124843	43374	107	107024	108462	45	Smiths Thinly Cut Roast Chicken 175g	
124845	43374	107	107024	108462	45	Smiths Thinly Cut Roast Chicken 175g	

Next steps:

[Generate code with](#) **duplicate_rows**

[New interactive sheet](#)

```
#dropping or deleting the duplicate values
transaction_data = transaction_data.drop_duplicates()
```

```
#One duplicate transaction value was identited and deleted to maintain data
```

```
purchase_behaviour_data.duplicated().sum()
#zero rows are identited as duplicated
```

```
np.int64(0)
```

```
transaction_data.describe()
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR
	264835.000000	264835.000000	2.648350e+05	2.648350e+05	264835.000000
count	43464.036600	135.080216	1.355496e+05	1.351584e+05	56.58320
mean	105.389336	76.784306	8.058011e+04	7.813316e+04	32.82669
std	43282.000000	1.000000	1.000000e+03	1.000000e+00	1.00000
min	43373.000000	70.000000	7.002100e+04	6.760100e+04	28.00000
25%	43464.000000	130.000000	1.303580e+05	1.351380e+05	56.00000
50%	43555.000000	203.000000	2.030945e+05	2.027015e+05	85.00000
75%	43646.000000	272.000000	2.373711e+06	2.415841e+06	114.00000
max					

```
transaction_data["PROD_QTY"].value_counts().sort_index()
```

	count
PROD_QTY	
1	27518
2	236038
3	430
4	397
5	450
200	2

dtype: int64

```
#identifying outliers in transaction dataset
transaction_data[transaction_data["PROD_QTY"]>5]
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_QTY
						Dorito	
69762	43331	226	226000	226201	4	Corn Chp Supreme 380g	20
						Dorito	
69763	43605	226	226000	226210	4	Corn Chp Supreme 380g	20

```
#Confirming that identified outlier is real
print(transaction_data[transaction_data['LYLTY_CARD_NBR']==226000])
```

```
print()
print(purchase_behaviour_data[purchase_behaviour_data['LYLTY_CARD_NBR']==226
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR \	
69762	43331	226	226000	226201	4	
69763	43605	226	226000	226210	4	

	PROD_NAME	PROD_QTY	TOT_SALES
69762	Dorito Corn Chp Supreme 380g	200	650.0
69763	Dorito Corn Chp Supreme 380g	200	650.0

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER
59694	226000	OLDER FAMILIES	Premium

```
#removing the outlier from transaction dataset
transaction_data = transaction_data[transaction_data['LYLTY_CARD_NBR'] != 22
```

```
#verifying that column is removed from the dataset or not.
transaction_data['LYLTY_CARD_NBR'].describe()
```

	LYLTY_CARD_NBR
count	1.355489e+05
mean	8.058003e+04
std	1.000000e+03
min	7.002100e+04
25%	1.303570e+05
50%	2.030940e+05
75%	2.373711e+06
max	

dtype: float64

```
transaction_data
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PR
						Natural Chip	
0	43390	1	1000	1	5	Compny SeaSalt175g	
						CCs Nacho Cheese 175g	
1	43599	1	1307	348	66	Smiths Crinkle Cut Chips	
						Chicken 170g	
2	43605	1	1343	383	61	Smiths Chip Thinly	
						S/Cream&Onion 175g	
3	43329	2	2373	974	69	Kettle Tortilla ChpsHny&Jlpno	
						Chili 150g	
4	43330	2	2426	1038	108	...	
...	...	...	...	...	...	Kettle Sweet Chilli And Sour Cream 175g	
264831	43533	272	272319	270088	89	Tostitos Splash Of Lime 175g	
264832	43325	272	272358	270154	74	Doritos Mexicana 170g	
264833	43410	272	272379	270187	51	Doritos Corn Chip Mexican Jalapeno 150g	
264834	43461	272	272379	270188	42	Tostitos Splash Of Lime 175g	
264835	43365	272	272380	270189	74		

264833 rows × 8 columns

```
transaction_data['DATE'].dtype
```

```
dtype('int64')
```

```
transaction_data["DATE"] = pd.to_datetime(
    transaction_data["DATE"],
    origin="1899-12-30",
    unit="D"
)
```

```
transaction_data.head()
```


	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_QTY
0	2018-10-17	1	1000	1	5	Natural Chip Comphy SeaSalt175g CCs Nacho Cheese 175g	2
1	2019-05-14	1	1307	348	66	Smiths Crinkle Cut Chips Chicken 170g	3
2	2019-05-20	1	1343	383	61	Smiths Chip Thinly S/Cream&Onion 175g	2
3	2018-08-17	2	2373	974	69	Kettle Tortilla ChpsHny&Jlpno Chili 150g	5
4	2018-08-18	2	2426	1038	108		3

```
#checking for the Maximum and Minimum date time in transaction dataset
print("The maximum date is : ",transaction_data['DATE'].max())
print("The minimum date is : ",transaction_data['DATE'].min())
```

```
The maximum date is : 2019-06-30 00:00:00
The minimum date is : 2018-07-01 00:00:00
```

```
# extracting Brand Name (first word)
transaction_data["BRAND_NAME"] = transaction_data["PROD_NAME"].str.extract(r
transaction_data
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PRO
						Natural Chip	
0	2018-10-17	1	1000	1	5	Comphy SeaSalt175g	
1	2019-05-14	1	1307	348	66	CCs Nacho Cheese 175g	
2	2019-05-20	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	2018-08-17	2	2373	974	69	Smiths Chip Thinly S/Cream&Onion 175g	
4	2018-08-18	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	
...	...	...	...	...	...	...	
264831	2019-03-09	272	272319	270088	89	Kettle Sweet Chilli And Sour Cream 175g	
264832	2018-08-13	272	272358	270154	74	Tostitos Splash Of Lime 175g	
264833	2018-11-06	272	272379	270187	51	Doritos Mexicana 170g	
264834	2018-12-27	272	272379	270188	42	Doritos Corn Chip Mexican Jalapeno 150g	
264835	2018-09-22	272	272380	270189	74	Tostitos Splash Of Lime 175g	

264833 rows × 9 columns

```
# Pack Size
pack_pattern = r'(\d+(?:\.\d+)?)\s?(?:kg|g|mg|l|ml|oz|lb|cl|pk|pcs|pc|ct))\b'

transaction_data["PACK_SIZE"] = (
    transaction_data["PROD_NAME"]
    .str.extract(pack_pattern, flags=re.IGNORECASE)[0]
    .str.replace(r"\s+", "", regex=True)
    .str.lower()
)
```

transaction_data

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PRO
						Natural Chip	
0	2018-10-17	1	1000	1	5	Comphny SeaSalt175g CCs Nacho Cheese 175g	
1	2019-05-14	1	1307	348	66	Smiths Crinkle Cut Chips Chicken 170g	
2	2019-05-20	1	1343	383	61	Smiths Chip Thinly S/Cream&Onion 175g	
3	2018-08-17	2	2373	974	69	Kettle Tortilla ChpsHny&Jlpno Chili 150g	
4	2018-08-18	2	2426	1038	108	...	
...	...	...	...	...	...	Kettle Sweet Chilli And Sour Cream 175g	
264831	2019-03-09	272	272319	270088	89	Tostitos Splash Of Lime 175g	
264832	2018-08-13	272	272358	270154	74	Doritos Mexicana 170g	
264833	2018-11-06	272	272379	270187	51	Doritos Corn Chip Mexican Jalapeno 150g	
264834	2018-12-27	272	272379	270188	42	Tostitos Splash Of Lime 175g	
264835	2018-09-22	272	272380	270189	74		

264833 rows × 10 columns

```
transaction_data['BRAND_NAME'].value_counts()
```

	count
BRAND_NAME	
Kettle	41288
Smiths	28859
Pringles	25102
Doritos	24962
Thins	14075
RRD	11894
Infuzions	11057
WW	10320
Cobs	9693
Tostitos	9471
Twisties	9454
Old	9324
Tyrrells	6442
Grain	6272
Natural	6050
Red	5885
Cheezels	4603
CCs	4551
Woolworths	4437
Dorito	3183
Infzns	3144
Smith	2963
Cheetos	2927
Snbts	1576
Burger	1564
GrnWves	1468
Sunbites	1432
NCC	1419
French	1418

dtype: int64

```
sorted(transaction_data['BRAND_NAME'].unique())
```

```
['Burger',  
'CCs',  
'Cheetos',  
'Cheezels',  
'Cobs',  
'Dorito',  
'Doritos',  
'French',  
'Grain',  
'GrnWves',  
'Infuzions',  
'Infzns',  
'Kettle',  
'NCC',  
'Natural',  
'Old',  
'Pringles',  
'RRD',  
'Red',  
'Smith',  
'Smiths',  
'Snbts',  
'Sunbites',  
'Thins',  
'Tostitos',  
'Twisties',  
'Tyrrells',  
'WW',  
'Woolworths']
```

```
unique_value = transaction_data.loc[transaction_data['BRAND_NAME'] == 'RRD']
```

```
unique_value
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_Q
92	2019-05-18	120	120226	123743	67	RRD Chilli& Coconut 150g	
118	2019-05-19	164	164183	164919	67	RRD Chilli& Coconut 150g	
140	2018-08-16	191	191200	192940	53	RRD Sweet Chilli & Sour Cream 165g	
294	2019-01-17	5	5018	4467	53	RRD Sweet Chilli & Sour Cream 165g	
302	2018-08-11	5	5028	4535	67	RRD Chilli& Coconut 150g	
...	...	...	...	...	...	RRD Steak & Chimuchurri 150g	
264752	2018-07-30	268	268365	264802	103	RRD Pc Sea Salt 165g	
264761	2019-06-21	269	269125	265778	11	RRD Chilli& Coconut 150g	
264778	2019-06-02	269	269204	266273	67	RRD Pc Sea Salt 165g	
264781	2019-02-08	269	269222	266383	11	RRD SR Slow Rst Pork Belly 150g	
264814	2018-08-30	271	271193	269364	10		

11894 rows × 10 columns

Next steps:

[Generate code with unique_value](#)[New interactive sheet](#)

```
print(transaction_data[
    transaction_data["BRAND_NAME"]=="Natural"
]["PROD_NAME"].unique())

print(transaction_data[
```

```
transaction_data["BRAND_NAME"]=="Old"
][["PROD_NAME"].unique())
```

```
['Old El Paso Salsa Dip Tomato Mild 300g'
'Old El Paso Salsa Dip Chnky Tom Ht300g'
'Old El Paso Salsa Dip Tomato Med 300g']

['Old El Paso Salsa Dip Tomato Mild 300g'
'Old El Paso Salsa Dip Chnky Tom Ht300g'
'Old El Paso Salsa Dip Tomato Med 300g']
```

```
#creating a mapping
brand_mapping = {
    'Infzns' : 'Infuzions',
    'Dorito' : 'Doritos',
    'Smith' : 'Smiths',
    'Grain' : 'GrnWves',
    'Red' : 'RRD',
    'WW' : 'Woolworths',
    'Snbts' : 'Sunbites',
    'Natural' : 'Natural Chip Co',
    'Old' : 'Old El Paso'
}
```

```
#standarizing the names
transaction_data["BRAND_NAME"] = transaction_data["BRAND_NAME"].replace(brand_mapping)
```

```
transaction_data
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PRO
						Natural Chip	
0	2018-10-17	1	1000	1	5	Comphy SeaSalt175g CCs Nacho Cheese 175g	
1	2019-05-14	1	1307	348	66	Smiths Crinkle Cut Chips Chicken 170g	
2	2019-05-20	1	1343	383	61	Smiths Chip Thinly S/Cream&Onion 175g	
3	2018-08-17	2	2373	974	69	Kettle Tortilla ChpsHny&Jlpno Chili 150g	
4	2018-08-18	2	2426	1038	108	...	
...	...	...	...	...	...	Kettle Sweet Chilli And Sour Cream 175g	
264831	2019-03-09	272	272319	270088	89	Tostitos Splash Of Lime 175g	
264832	2018-08-13	272	272358	270154	74	Doritos Mexicana 170g	
264833	2018-11-06	272	272379	270187	51	Doritos Corn Chip Mexican Jalapeno 150g	
264834	2018-12-27	272	272379	270188	42	Tostitos Splash Of Lime 175g	
264835	2018-09-22	272	272380	270189	74		

264833 rows × 10 columns

```
transaction_data.to_excel("Final_transaction_data.xlsx",index=False)
```

```
#Merging Final transaction dataset with purchase behaviour dataset
merged_data = pd.merge(
    transaction_data,
    purchase_behaviour_data,
    on="LYLTY_CARD_NBR",
    how="left"
)
```

```
merged_data.shape
```

```
(264833, 12)
```



```
merged_data.isnull().sum()
```

	0
DATE	0
STORE_NBR	0
LYLTY_CARD_NBR	0
TXN_ID	0
PROD_NBR	0
PROD_NAME	0
PROD_QTY	0
TOT_SALES	0
BRAND_NAME	0
PACK_SIZE	0
LIFESTAGE	0
PREMIUM_CUSTOMER	0

dtype: int64

```
merged_data.head()
```

OD_NBR	PROD_NAME	PROD_QTY
	Natural Chip	
	Compny	
5	SeaSalt175g	2
	CCs Nacho	
	Cheese 175g	
66	Smiths Crinkle	3
	Cut Chips	
	Chicken 170g	
61	Smiths Chip	2
	Thinly	
	S/Cream&Onion	
69	175g	5
	Kettle Tortilla	
	ChpsHny&Jlpno	
108	Chili 150g	3

```
#adding a metrics (Price Per Unit)
merged_data["PRICE_PER_UNIT"] = (
```

```
merged_data["TOT_SALES"] / merged_data["PROD_QTY"]  
)
```

```
merged_data
```

D_NBR	PROD_NAME	PRO
	Naturar Chip	
	Compny	
5	SeaSalt175g	
	CCs Nacho	
	Cheese 175g	
66	Smiths Crinkle	
	Cut Chips	
	Chicken 170g	
61	Smiths Chip	
	Thinly	
	S/Cream&Onion	
69	175g	
	Kettle Tortilla	
	ChpsHny&Jlpno	
108	Chili 150g	
	...	
...	Kettle Sweet	
	Chilli And Sour	
89	Cream 175g	
	Tostitos Splash	
	Of Lime 175g	
74		
	Doritos	
	Mexicana 170g	
51		
	Doritos Corn	
	Chip Mexican	
42	Jalapeno 150g	
	Tostitos Splash	
74	Of Lime 175g	

```
#which lifestage has generated more revenue  
merged_data.groupby("LIFESTAGE")["TOT_SALES"].sum().sort_values(ascending=Fa
```

TOT_SALES	
LIFESTAGE	
OLDER SINGLES/COUPLES	402420.75
RETIREEES	366470.90
OLDER FAMILIES	352467.20
YOUNG FAMILIES	316160.10
YOUNG SINGLES/COUPLES	260405.30
MIDAGE SINGLES/COUPLES	184751.30
NEW FAMILIES	50433.45

dtype: float64

```
#which customer category has generated more sales  
merged_data.groupby("PREMIUM_CUSTOMER")["TOT_SALES"].sum()
```

TOT_SALES	
PREMIUM_CUSTOMER	
Budget	676211.55
Mainstream	750744.50
Premium	506152.95

dtype: float64

```
#who has generated more sales in both category  
merged_data.groupby(  
    ["LIFESTAGE", "PREMIUM_CUSTOMER"]  
)["TOT_SALES"].sum()
```

		TOT_SALES
LIFESTAGE	PREMIUM_CUSTOMER	
MIDAGE SINGLES/COUPLES	Budget	35514.80
	Mainstream	90803.85
	Premium	58432.65
NEW FAMILIES	Budget	21928.45
	Mainstream	17013.90
	Premium	11491.10
OLDER FAMILIES	Budget	168363.25
	Mainstream	103445.55
	Premium	80658.40
OLDER SINGLES/COUPLES	Budget	136769.80
	Mainstream	133393.80
	Premium	132257.15
RETIREEES	Budget	113147.80
	Mainstream	155677.05
	Premium	97646.05
YOUNG FAMILIES	Budget	139345.85
	Mainstream	92788.75
	Premium	84025.50
YOUNG SINGLES/COUPLES	Budget	61141.60
	Mainstream	157621.60
	Premium	41642.10

dtype: float64

```
# unique customers in each segments
merged_data.groupby(
    ["LIFESTAGE", "PREMIUM_CUSTOMER"]
)["LYLTY_CARD_NBR"].nunique()
```

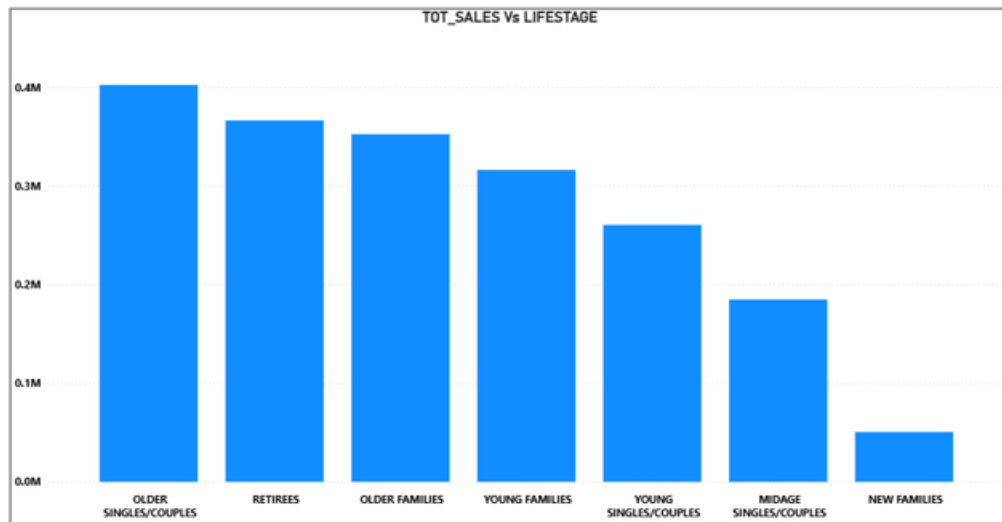
		LYLTY_CARD_NBR
LIFESTAGE	PREMIUM_CUSTOMER	
MIDAGE SINGLES/COUPLES	Budget	1504
	Mainstream	3340
	Premium	2431
NEW FAMILIES	Budget	1112
	Mainstream	849
	Premium	588
OLDER FAMILIES	Budget	4675
	Mainstream	2831
	Premium	2273
OLDER SINGLES/COUPLES	Budget	4929
	Mainstream	4930
	Premium	4750
RETIREEES	Budget	4454
	Mainstream	6479
	Premium	3872
YOUNG FAMILIES	Budget	4017
	Mainstream	2728
	Premium	2433
YOUNG SINGLES/COUPLES	Budget	3779
	Mainstream	8088
	Premium	2574

dtype: int64

```
#average spend per customers
merged_data.groupby(
    ['LIFESTAGE', 'PREMIUM_CUSTOMER']
)['TOT_SALES'].mean()
```

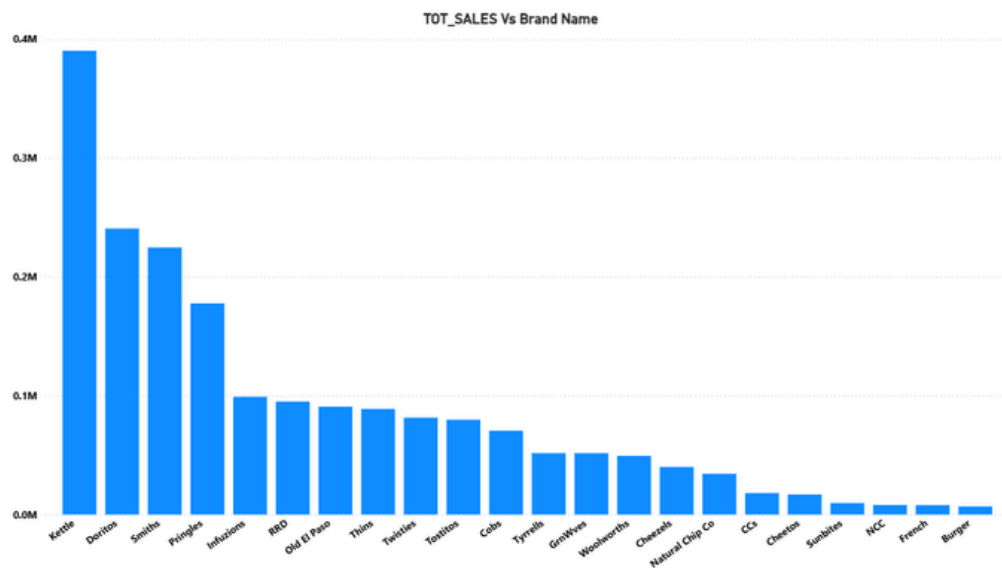
		TOT_SALES
LIFESTAGE	PREMIUM_CUSTOMER	
MIDAGE SINGLES/COUPLES	Budget	7.074661
	Mainstream	7.647284
	Premium	7.112056
NEW FAMILIES	Budget	7.297321
	Mainstream	7.317806
	Premium	7.231655
OLDER FAMILIES	Budget	7.269570
	Mainstream	7.262395
	Premium	7.208079
OLDER SINGLES/COUPLES	Budget	7.430315
	Mainstream	7.282116
	Premium	7.449848
RETIREEES	Budget	7.443445
	Mainstream	7.252262
	Premium	7.456174
YOUNG FAMILIES	Budget	7.287201
	Mainstream	7.189025
	Premium	7.266756
#XvOeUraNgGe SqINuaGnLtEitSy/C pOubhLaEsSed		6.615624
merged_data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'],		7.558339
)['PROD_QTY'].mean()		6.629852
		Premium
		PROD_QTY
dtype: float64		
LIFESTAGE	PREMIUM_CUSTOMER	
MIDAGE SINGLES/COUPLES	Budget	1.891633
	Mainstream	1.911656
	Premium	1.889727
NEW FAMILIES	Budget	1.853910
	Mainstream	
	Premium	

1. Sales Vs Life Stage



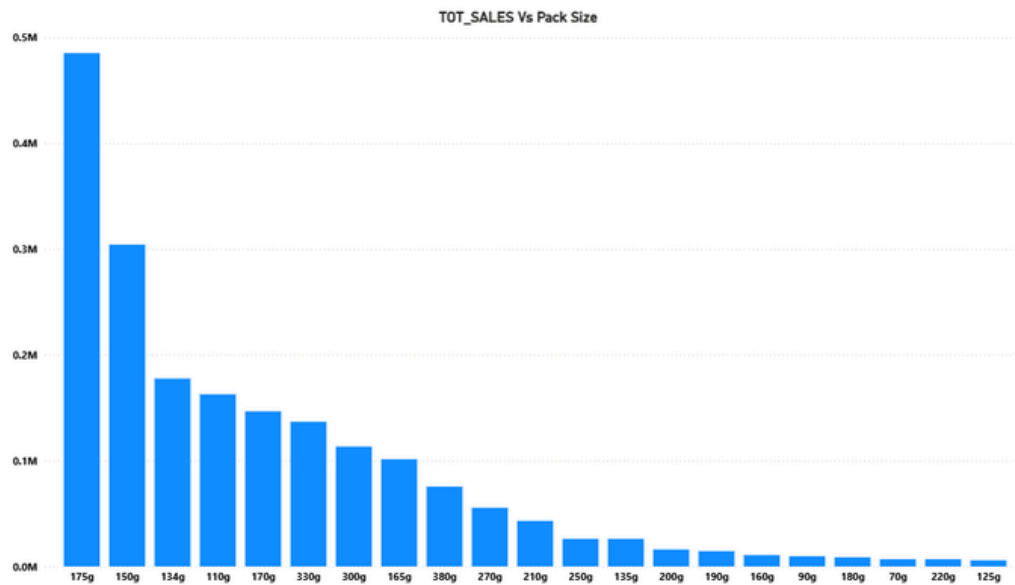
Older Families contribute the highest chip sales, indicating that families are a key revenue-generating customer segment.

2. Sales Vs Brand



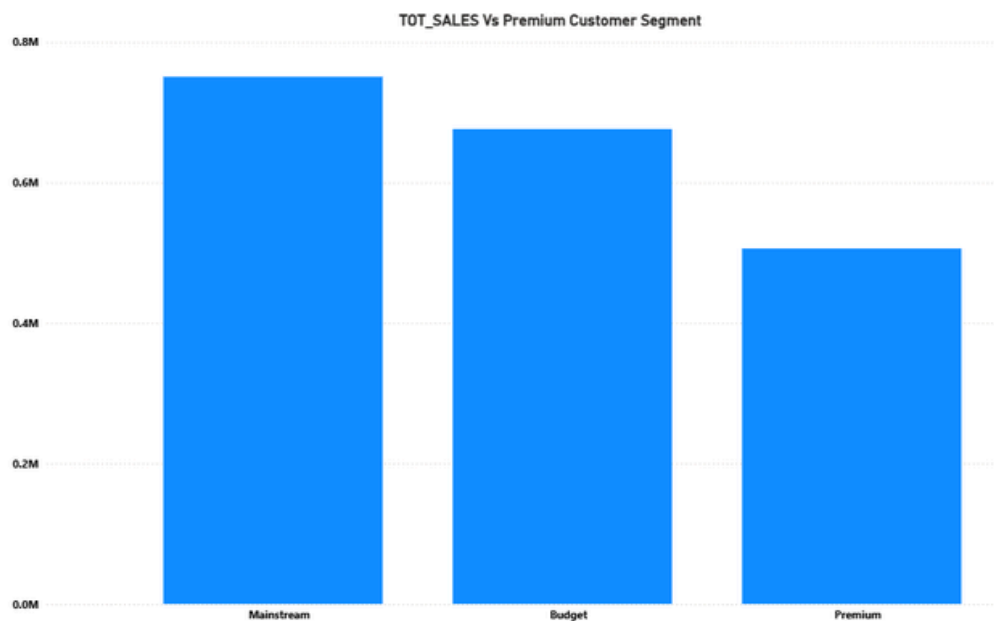
Kettle and Smiths are among the top-selling brands, suggesting strong customer preference for these products.

3. Sales Vs Pack Size



Medium-sized packs (e.g., 175g) account for the highest sales, indicating they are the preferred pack size.

4. Sales Vs Premium Customer Segment



Budget and Mainstream customers contribute a significant share of total sales, making them important target segments.